

Time: Three Hours

Maximum Marks: 75

SECTION - A ($10 \times 2 = 20$ Marks)**Answer Any TEN Questions**

1. State Binomial theorem.
2. If $t_n = \frac{x^n}{n!}$ then find the value $\lim_{n \rightarrow \infty} t_n$.
3. Write down the expansion of $\log(1+x)$.
4. Write down the condition for skew-matrix.
5. Define skew Hermitian matrix.
6. State Cayley Hamilton theorem.
7. Write down Newton's method formula.
8. Write down Newton's forward interpolation formula.
9. Write down the expansion of $\tan n\theta$.
10. Write down the formula for $\tan h(x+y)$.
11. If $y = \frac{x+1}{x+2}$ then find $\frac{dy}{dx}$.
12. Find $D^n[f(x)]$ if $f(x)$ is $\sin(ax+b)$.

SECTION - B ($5 \times 5 = 25$ Marks)**Answer Any FIVE Questions**

13. Write down the middle term of $\left(x - \frac{2}{x}\right)^{12}$.
14. Show that $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ is orthogonal.

15. Using Lagrange's interpolation formula, find the value of Y corresponding to $x = 10$ from the following data.

x	5	6	9	11
y	12	13	14	16

16. Find the positive root of $x^4 - x - 10 = 0$ near $x = 2$ by Newton's method, correct to four decimal places.
17. Express $\sin 7\theta$ in terms of $\sin \theta$.
18. Prove that $\sin^5 \theta = \frac{1}{16} [\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta]$.
19. Find the value of the $\frac{\partial(u,v)}{\partial(r,\theta)}$ where $u = x^2 - y^2$, $v = 2xy$ and $x = r \cos \theta$, $y = r \sin \theta$.

SECTION - C ($3 \times 10 = 30$ Marks)**Answer Any THREE Questions**

20. Show that $\frac{1}{n+1} + \frac{1}{2(n+1)^2} + \dots = \frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \dots$.
21. Verify Cayley-Hamilton theorem and hence find inverse for $\begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix}$.
22. Using Newton's backward interpolation formula, find y when $x = 27$ from the following data.

x	10	15	20	25	30
y	35.4	32.2	29.1	26.0	23.1

23. If $\tan(\theta + i\phi) = \cos \alpha + i \sin \alpha$ then prove that $\theta = \frac{n\pi}{2} + \frac{\pi}{4}$ and $\phi = \frac{1}{2} \log \tan \left(\frac{\pi}{4} + \frac{\alpha}{2} \right)$.
24. If $u = \sin^{-1} \left(\frac{y}{x} \right)$ then prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$.